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AI work, March to September 2026

I'm a second-year electrical engineering student at Waterloo. In March I started using AI coding agents seriously, on my own. Six months later a server at home runs most of my projects: agents plan the work, write it, check it and ship it, and I steer from my phone. Along the way I built an agent that designs circuit boards and a tool that builds interactive lessons for my courses. I wrote the rules the agents' work has to pass. The agents wrote most of the code. My co-ops were in hardware, at Arista and Fab2, and I'm looking for a January to April 2027 co-op.

550

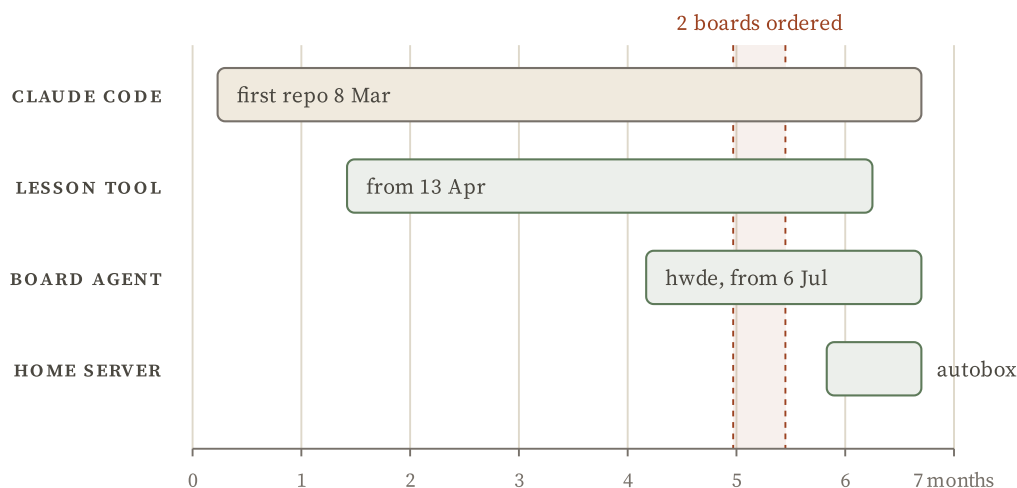
pieces of work landed by the server in under four weeks

16

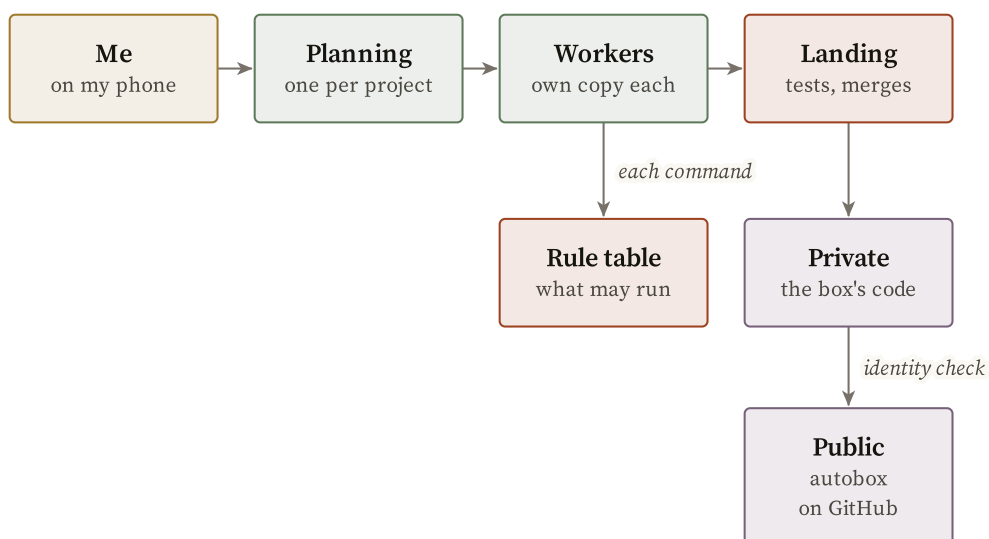
boards designed by the board agent, 2 of them made

12/12

planted mistakes caught by the lesson reviewer



Months since 1 March 2026. I dropped a first board-agent prototype in June, two days after it worked, and started again properly as hwde.



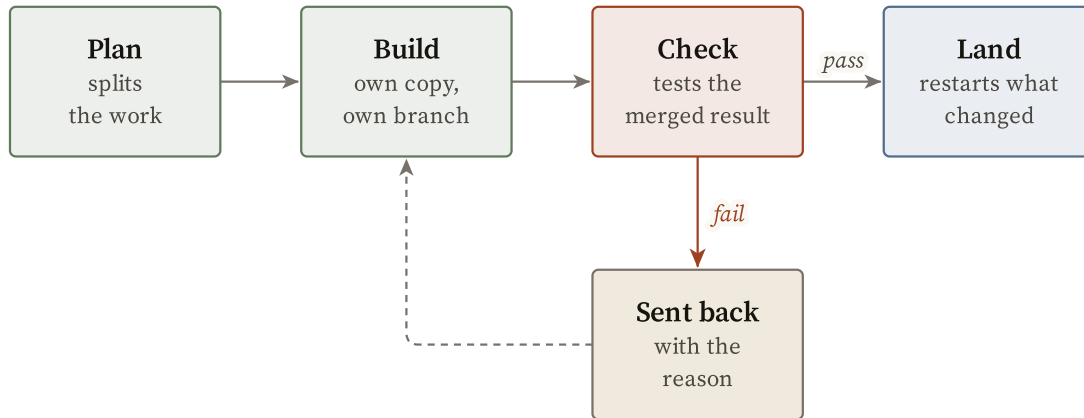
How my projects run now, on the server described next. Every change goes through the landing, and every command an agent runs goes past the rule table first.

autobox

github.com/ihsan-sa/autobox

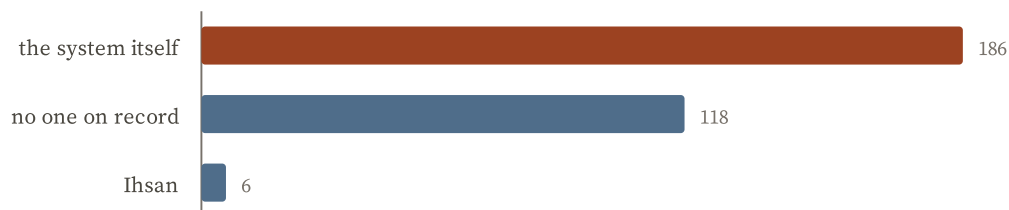
A server at home where AI agents do the work on my projects and check it before it's accepted.

Read more: [why it's built this way](#) · [what it does and how to run it](#)



What happens to one change.

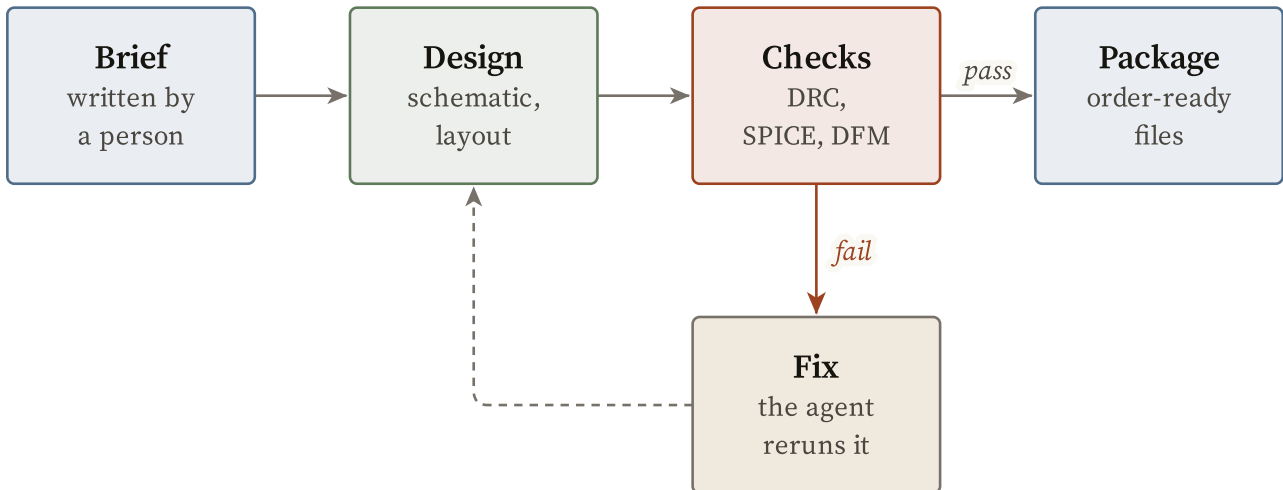
- **550 pieces of work landed** (pull requests) from 27 August to 20 September 2026. Each agent works on its own copy of the project, and nothing merges until the tests pass on the change combined with the latest main code. Merge queues do this for people; I applied it to work nobody reads first.
- **A rule table decides what an agent may run** before it runs it. After one fix I replayed 24,013 commands the agents had really run: the new rule would have blocked 1.8% of them.
- **The public repo can't leak the private half.** A gate refuses to publish if my identity shows up in the change.
- **It writes down everything that goes wrong, and who noticed.**



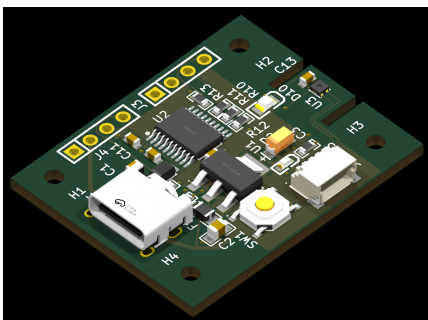
310 logged problems, by who noticed them. Most of the 118 with no one on record were typed up after the fact, when that field often gets left blank.

You describe a circuit board in plain English, and an agent produces the files a factory needs.

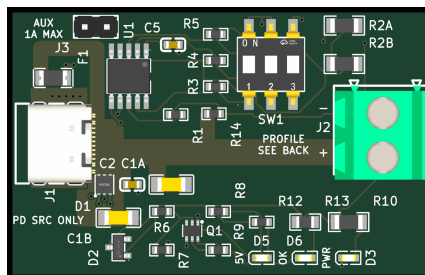
Read more: [how the pipeline works](#) · [the 13.6-hour run, step by step](#) · [the planted faults](#)



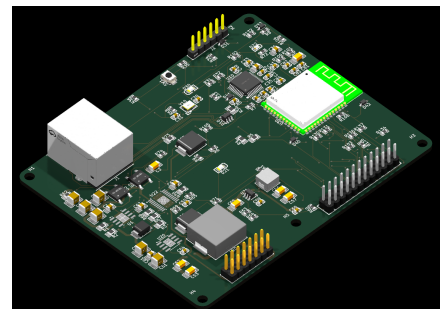
- **Checked at every step by the tools an engineer uses:** KiCad's design-rule check, the ngspice circuit simulator and the Freerouting auto-router. How well the router does is fed back into where parts get placed.
- **16 boards designed, 6 ready to order, 2 made.** Neither made board has been powered on yet.
- **Best run: brief to order-ready board in 13.6 hours, with nobody watching.** 26 parts, 3 rounds, zero design-rule violations. Ordering always waits for a person to type a confirmation.
- **The checks are tested too.** 3 good boards and 7 with planted faults, like a missing via, each tied to the check that must catch it. Reading the code, I found the agent could approve its own failed check; the fix is to take approval out of its reach.



g0-sense, the 13.6-hour run



pd-trigger, made



lumina-carrier, 4 layers, made

KiCad renders of three boards the agent designed. The first was never ordered.

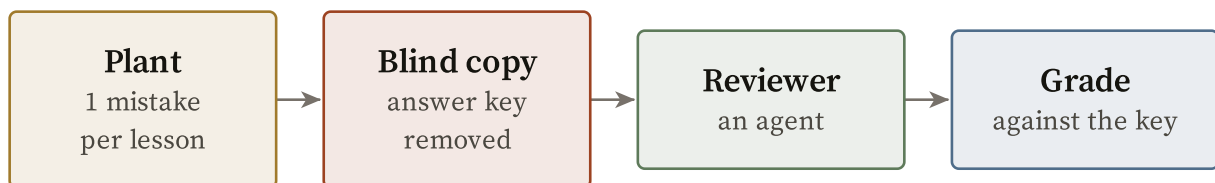
lesson-builder

github.com/ihsan-sa/lesson-builder

A Claude Code skill that builds interactive lessons for my courses, with a tutor inside each one.

Read more: [how it builds a lesson](#) · [the blind reviewer test](#)

- **41 lessons built across 8 courses**, mostly my own at Waterloo, by 12 agents working in six phases.
- **The tutor can rewrite the lesson** while the student is reading it.
- **I tested its reviewer blind.** I planted one mistake in each of 12 lessons and took the answer key away before it looked.



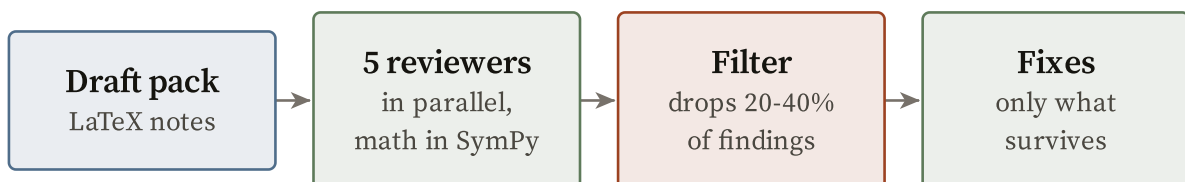
It caught 12 of 12, and it found 2 real flaws in text I meant to be clean. It's the only recorded time one of my checks caught something I didn't plant.

pdf-material-builder

github.com/ihsan-sa/pdf-material-builder

A skill that turns a course into a set of LaTeX study notes.

Read more: [the five reviewers](#) and [the filter](#)

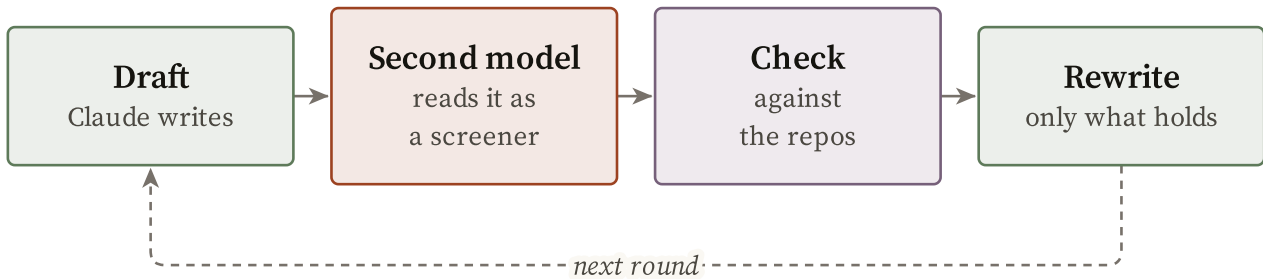


- **Five reviewers read every pack at once:** one for content, one playing a brutal student, one for consistency across documents, one that checks the math in SymPy, and a reader seeing it cold.
- **A filter then throws out 20 to 40% of their findings**, using a list of each course's past false alarms.
- Built real notes for three courses.

A second opinion on my resume

not public

Before anything goes out with my name on it, a model from another lab reads it cold, and every claim is checked against the repos.



- **It caught me calling one idea mine.** Testing the merged result is what merge queues have done for years, so that claim came out.
- **It pointed me at a published 679-board benchmark** for board-layout agents, so hwde is now described next to it.
- **Every number on my resume has its exact value and source file** written down beside it. This document works the same way.

Not done yet

No board from hwde has been powered on. I haven't yet measured how many of the planted faults its checks catch that KiCad's own check misses. Those are the next two things on my list.